Lecture 15: Spectral Decomposition

MATH203 Applied Linear Algebra

Why Do We Need A^n?

What Is a_{100}?

$$a_0 = 0, \qquad a_1 = 1, \qquad a_{n+1} = 3a_n - 2a_{n-1}$$

What Is a_{100}?

$$a_0 = 0, \qquad a_1 = 1, \qquad a_{n+1} = 3a_n - 2a_{n-1}$$

The first few terms: $0,\ 1,\ 3,\ 7,\ 15,\ 31,\ \ldots$

What Is a_{100}?

$$a_0 = 0, \qquad a_1 = 1, \qquad a_{n+1} = 3a_n - 2a_{n-1}$$

The first few terms: $0,\ 1,\ 3,\ 7,\ 15,\ 31,\ \ldots$

Rewrite as a matrix recurrence:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \underbrace{\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}}_{B} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix}$$

After n steps:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = B^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

To answer the question, we need B^{100}.

A^n Is Everywhere

Probability. Flip a coin repeatedly. Heads $\to$ walk **3** meters. Tails $\to$ walk **2** meters.

What is the probability of landing **exactly** on the n-meter mark?

$$p_n = \frac{1}{2}\, p_{n-2} + \frac{1}{2}\, p_{n-3} \qquad \longrightarrow \qquad \text{need } A^n$$

A^n Is Everywhere

Probability. Flip a coin repeatedly. Heads $\to$ walk **3** meters. Tails $\to$ walk **2** meters.

What is the probability of landing **exactly** on the n-meter mark?

$$p_n = \tfrac{1}{2}\, p_{n-2} + \tfrac{1}{2}\, p_{n-3} \qquad \longrightarrow \qquad \text{need } A^n$$

Money sharing. Three cousins each start with some money.

Each round: pool half your money, split the pool equally.

$$\vec{m}_n = \underbrace{\begin{pmatrix} 2/3 & 1/6 & 1/6 \\ 1/6 & 2/3 & 1/6 \\ 1/6 & 1/6 & 2/3 \end{pmatrix}}_{M} \vec{m}_{n-1} \qquad \longrightarrow \qquad \text{need } M^n$$

What happens as $n \to \infty$? Does everyone end up with the same amount?

Lecture 14's Method: One Step at a Time

Lecture 14 taught us: $A^2 = 5A + 2I$ for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

$$
\begin{aligned}
A^3 &= A \cdot A^2 = A(5A + 2I) = 5A^2 + 2A = 27A + 10I \\
A^4 &= A \cdot A^3 = 145A + 54I \\
A^5 &= 781A + 290I \\
&\vdots
\end{aligned}
$$

Lecture 14's Method: One Step at a Time

Lecture 14 taught us: $A^2 = 5A + 2I$ for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

$$
\begin{aligned}
A^3 &= A \cdot A^2 = A(5A + 2I) = 5A^2 + 2A = 27A + 10I \\
A^4 &= A \cdot A^3 = 145A + 54I \\
A^5 &= 781A + 290I \\
&\vdots
\end{aligned}
$$

Each power needs **the previous power**.

Want A^{100}? Repeat **99 times**.

This is not a formula. This is bookkeeping.

Can we find a **closed formula** for A^n?

Just plug in n — get the answer immediately?

The Dream: Projections Make Powers Trivial

From Lectures 7–10, you know **projections**:

$$P^2 = P \implies P^n = P \quad \text{for all } n \geq 1.$$

The Dream: Projections Make Powers Trivial

From Lectures 7–10, you know **projections**:

$$P^2 = P \implies P^n = P \quad \text{for all } n \geq 1.$$

What if we could decompose A into projections?

$$A = \underbrace{2}_{\lambda_1} \cdot P_1 + \underbrace{3}_{\lambda_2} \cdot P_2$$

Then:

$$A^n = 2^n \cdot \underbrace{P_1^n}_{=P_1} + 3^n \cdot \underbrace{P_2^n}_{=P_2} = \boxed{2^n P_1 + 3^n P_2}$$

The Dream: Projections Make Powers Trivial

From Lectures 7–10, you know **projections**:

$$P^2 = P \implies P^n = P \quad \text{for all } n \geq 1.$$

What if we could decompose A into projections?

$$A = \underbrace{2}_{\lambda_1} \cdot P_1 + \underbrace{3}_{\lambda_2} \cdot P_2$$

Then:

$$A^n = 2^n \cdot \underbrace{P_1^n}_{=P_1} + 3^n \cdot \underbrace{P_2^n}_{=P_2} = \boxed{2^n P_1 + 3^n P_2}$$

A **closed formula**. Plug in $n = 100$? Done.

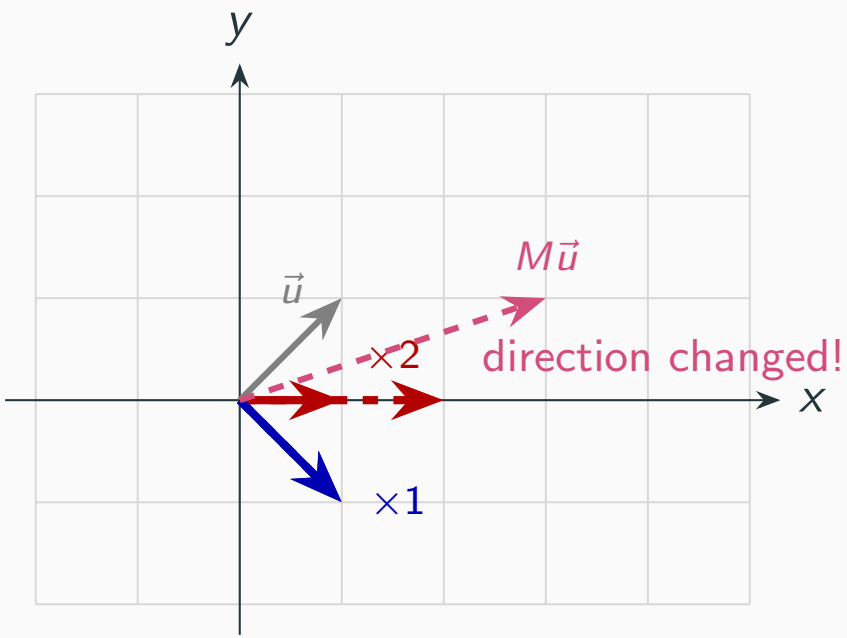
Slogan: Projections are the simplest matrices.

To understand any matrix, decompose it into projections.

Eigenvalues and Eigenvectors

Some Vectors Are Special

Multiply three vectors by $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$:



$\vec{v}_1 = (1, 0)$: stays on the same line, stretched by 2.

$\vec{v}_2 = (1, -1)$: stays on the same line, unchanged.

These vectors just get **scaled**. No rotation, no shearing.

Definition: Eigenvalue and Eigenvector

Definition

A **non-zero** vector $\vec{v}$ is an **eigenvector** of A for **eigenvalue** λ if

$$A\vec{v} = \lambda\vec{v}.$$

In words: A acts on $\vec{v}$ by just **scaling**.

$\lambda > 1$	vector gets **stretched**
$0 < \lambda < 1$	vector gets **shrunk**
$\lambda = 0$	vector gets **killed** (sent to $\vec{0}$)
$\lambda < 0$	vector gets **flipped** and scaled
$\lambda = 1$	vector **doesn't move**

Where Do Eigenvalues Come From?

From Lecture 14: the **characteristic polynomial**

$$\det(tI - A) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_n)$$

The roots $\lambda_1, \ldots, \lambda_n$ are the **eigenvalues**.

Running example for today:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \det(tI - A) = t(t-1)(t-2)$$

Eigenvalues: $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = 2$.

What Lecture 14 Also Told Us

Recall the **Cayley–Hamilton Theorem**:

the characteristic polynomial $\det(tI - A)$ **annihilates** A.

For our example: $\det(tI - A) = t(t-1)(t-2)$.

Plug in $t = A$:

$$\underbrace{A(A-I)(A-2I)}_{\text{product of } (A-\lambda_i I) \text{ over all eigenvalues}} = 0$$

What Lecture 14 Also Told Us

Recall the **Cayley–Hamilton Theorem**:

the characteristic polynomial $\det(tI - A)$ **annihilates** A.

For our example: $\det(tI - A) = t(t-1)(t-2)$.

Plug in $t = A$:

$$\underbrace{A(A-I)(A-2I)}_{\text{product of } (A-\lambda_i I) \text{ over all eigenvalues}} = 0$$

The eigenvalues give us an **annihilating polynomial that factors into linear terms**.

This factored form is the key to everything that follows.

Why This Changes Everything

Take **any** polynomial $g(t)$. Divide by the annihilating polynomial:

$$g(t) = Q(t) \cdot \underbrace{t(t-1)(t-2)}_{\det(tI-A)} + \underbrace{R(t)}_{\deg<3}$$

Why This Changes Everything

Take **any** polynomial $g(t)$. Divide by the annihilating polynomial:

$$g(t) = Q(t) \cdot \underbrace{t(t-1)(t-2)}_{\det(tI-A)} + \underbrace{R(t)}_{\deg<3}$$

Plug in $t = A$:

$$g(A) = Q(A) \cdot \underbrace{A(A-I)(A-2I)}_{= 0 \text{ by Cayley–Hamilton!}} + R(A) = R(A)$$

Why This Changes Everything

Take **any** polynomial $g(t)$. Divide by the annihilating polynomial:

$$g(t) = Q(t) \cdot \underbrace{t(t-1)(t-2)}_{\det(tI-A)} + \underbrace{R(t)}_{\deg<3}$$

Plug in $t = A$:

$$g(A) = Q(A) \cdot \underbrace{A(A-I)(A-2I)}_{= 0 \text{ by Cayley–Hamilton!}} + R(A) = R(A)$$

So $g(A)$ depends only on the **remainder** $R(t)$.

$R(t)$ has degree < 3, so it is determined by **three values**:

$$g(0), \quad g(1), \quad g(2) \qquad \text{(the values at the eigenvalues!)}$$

If we can express R in terms of these values, we get:

$$g(A) = g(0) \cdot P_0 + g(1) \cdot P_1 + g(2) \cdot P_2$$

for some matrices P_0, P_1, P_2. **The dream from §1!**

Lecture 14's Remainder Theorem divides by **one** factor.

We need to divide by **three** factors at once.

How?

From Remainder Theorem to Lagrange Interpolation

$m = 1$: The Remainder Theorem (Lecture 14)

Start with what we know. Dividing $g(t)$ by **one** factor $(t - a)$:

$$g(t) = Q(t) \cdot (t - a) + \underbrace{g(a)}_{\text{remainder}}$$

$m = 1$: The Remainder Theorem (Lecture 14)

Start with what we know. Dividing $g(t)$ by **one** factor $(t - a)$:

$$g(t) = Q(t) \cdot (t - a) + \underbrace{g(a)}_{\text{remainder}}$$

Rewrite:

$$g(t) = Q(t) \cdot (t - a) + g(a) \cdot \underbrace{1}_{f_a(t)}$$

The "Lagrange basis" for **1 point** is just $f_a(t) = 1$:

	$f_a(t) = 1$
$t = a$	1

One point $\rightarrow$ one value determines the remainder.

$m = 2$: Two Points

Dividing $g(t)$ by $(t-1)(t-2)$: remainder has degree ≤ 1, determined by **two** values: $g(1)$ and $g(2)$.

Need **basis polynomials** that pick out each value:

	$\underbrace{-(t-2)}_{f_1(t)}$	$\underbrace{(t-1)}_{f_2(t)}$
$t = 1$	1	0
$t = 2$	0	1

$$g(t) = Q(t)(t-1)(t-2) + g(1) \cdot f_1(t) + g(2) \cdot f_2(t)$$

Same shape as $m = 1$, but with **2 points** instead of 1.

$m = 3$: Three Points — at $0, 1, 2$

Dividing $g(t)$ by $t(t-1)(t-2)$: remainder has degree ≤ 2,

determined by **three** values: $g(0)$, $g(1)$, $g(2)$.

	$\underbrace{\frac{(t-1)(t-2)}{(0-1)(0-2)}}_{f_0(t)=\frac{(t-1)(t-2)}{2}}$	$\underbrace{\frac{t(t-2)}{(1-0)(1-2)}}_{f_1(t)=-t(t-2)}$	$\underbrace{\frac{t(t-1)}{(2-0)(2-1)}}_{f_2(t)=\frac{t(t-1)}{2}}$
$t = 0$	1	0	0
$t = 1$	0	1	0
$t = 2$	0	0	1

$$g(t) = Q(t) \cdot t(t-1)(t-2) + g(0) \cdot f_0(t) + g(1) \cdot f_1(t) + g(2) \cdot f_2(t)$$

Pattern: $f_{\lambda_i}(t) = \dfrac{\text{product of all } (t - \lambda_j) \text{ for } j \neq i}{\text{same product evaluated at } \lambda_i}$

Why Does Lagrange Always Work?

Polynomials of degree $< m$ form a **vector space** of dimension m.

Natural basis: $\{1,\ t,\ t^2,\ \ldots,\ t^{m-1}\}$ (m vectors)

Why Does Lagrange Always Work?

Polynomials of degree $< m$ form a **vector space** of dimension m.

Natural basis: $\{1,\ t,\ t^2,\ \ldots,\ t^{m-1}\}$ (m vectors)

The Lagrange polynomials $f_{\lambda_1}(t), \ldots, f_{\lambda_m}(t)$ are **also** m polynomials of degree $< m$.

Are they linearly independent? Look at their value tables:

	λ_1	λ_2	$\cdots$	λ_m
f_{λ_1}	**1**	0	$\cdots$	0
f_{λ_2}	0	**1**	$\cdots$	0
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
f_{λ_m}	0	0	$\cdots$	**1**

The value table is the **identity matrix** — so they are **linearly independent**.

Why Does Lagrange Always Work?

Polynomials of degree $< m$ form a **vector space** of dimension m.

Natural basis: $\{1,\ t,\ t^2,\ \ldots,\ t^{m-1}\}$ (m vectors)

The Lagrange polynomials $f_{\lambda_1}(t), \ldots, f_{\lambda_m}(t)$ are **also** m polynomials of degree $< m$.

Are they linearly independent? Look at their value tables:

	λ_1	λ_2	$\cdots$	λ_m
f_{λ_1}	**1**	0	$\cdots$	0
f_{λ_2}	0	**1**	$\cdots$	0
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
f_{λ_m}	0	0	$\cdots$	**1**

The value table is the **identity matrix** — so they are **linearly independent**.

m independent vectors in an m-dimensional space $\implies$ they form a **basis**.

Every polynomial of degree $< m$ is a unique linear combination of the Lagrange basis.

Two bases for the same space:

$$\underbrace{1,\ t,\ \ldots,\ t^{m-1}}_{\text{natural basis}} \quad \longleftrightarrow \quad \underbrace{f_{\lambda_1},\ f_{\lambda_2},\ \ldots,\ f_{\lambda_m}}_{\text{Lagrange basis}}$$

Lagrange Interpolation = Generalized Remainder Theorem

Theorem (Lagrange Interpolation)

For any polynomial $g(t)$ and distinct points $\lambda_1, \ldots, \lambda_m$:

$$g(t) = Q(t)(t-\lambda_1)\cdots(t-\lambda_m) + g(\lambda_1)f_{\lambda_1}(t) + \cdots + g(\lambda_m)f_{\lambda_m}(t)$$

Dividing by	Remainder determined by	Name
$(t-a)$	**1** value: $g(a)$	Remainder Thm (L14)
$(t-a)(t-b)$	**2** values: $g(a)$, $g(b)$	Lagrange, 2 pts
$(t-\lambda_1)\cdots(t-\lambda_m)$	m values: $g(\lambda_1), \ldots, g(\lambda_m)$	Lagrange, m pts

The Remainder Theorem is the **special case** $m = 1$.

Spectral Decomposition

Lagrange gives us a formula for $g(t)$.

Cayley-Hamilton says $A(A - I)(A - 2I) = 0$.

What happens if we plug in $t = A$?

The Trick: Plug In A

Apply Lagrange to $g(t) = t^n$ at the eigenvalues $0, 1, 2$:

$$t^n = Q(t) \cdot \underbrace{t(t-1)(t-2)}_{\det(tI-A)} + 0^n f_0(t) + 1^n f_1(t) + 2^n f_2(t)$$

The Trick: Plug In A

Apply Lagrange to $g(t) = t^n$ at the eigenvalues $0, 1, 2$:

$$t^n = Q(t) \cdot \underbrace{t(t-1)(t-2)}_{\det(tI-A)} + 0^n f_0(t) + 1^n f_1(t) + 2^n f_2(t)$$

Now substitute $t = A$:

$$A^n = Q(A) \cdot \underbrace{A(A-I)(A-2I)}_{= 0 \text{ by Cayley–Hamilton!}} + 0^n f_0(A) + 1^n f_1(A) + 2^n f_2(A)$$

The Trick: Plug In A

Apply Lagrange to $g(t) = t^n$ at the eigenvalues $0, 1, 2$:

$$t^n = Q(t) \cdot \underbrace{t(t-1)(t-2)}_{\det(tI-A)} + 0^n f_0(t) + 1^n f_1(t) + 2^n f_2(t)$$

Now substitute $t = A$:

$$A^n = Q(A) \cdot \underbrace{A(A-I)(A-2I)}_{= 0 \text{ by Cayley–Hamilton!}} + 0^n f_0(A) + 1^n f_1(A) + 2^n f_2(A)$$

The quotient term **vanishes**! For $n \geq 1$:

$$\boxed{A^n = f_1(A) + 2^n \cdot f_2(A)}$$

Closed formula. Just plug in n.

Compute A^n Explicitly

$$f_1(A) = -A(A - 2I) = -\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Compute A^n Explicitly

$$f_1(A) = -A(A-2I) = -\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$f_2(A) = \frac{A(A-I)}{2} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Compute A^n Explicitly

$$f_1(A) = -A(A-2I) = -\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$f_2(A) = \frac{A(A-I)}{2} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

For $n \geq 1$:

$$\boxed{A^n = \begin{pmatrix} 2^{n-1} & 0 & 2^{n-1} \\ 0 & 1 & 0 \\ 2^{n-1} & 0 & 2^{n-1} \end{pmatrix}}$$

The General Principle: $g(A)$ from Eigenvalues

Theorem (Spectral Substitution)

If $\det(tI - A) = (t - \lambda_1) \cdots (t - \lambda_m)$ with **distinct** λ_i, then:

$$g(A) = g(\lambda_1)\underbrace{f_{\lambda_1}(A)}_{P_{\lambda_1}} + g(\lambda_2)\underbrace{f_{\lambda_2}(A)}_{P_{\lambda_2}} + \cdots + g(\lambda_m)\underbrace{f_{\lambda_m}(A)}_{P_{\lambda_m}}$$

$g(A)$ is completely determined by its **value table**:

	λ_1	λ_2	$\cdots$	λ_m
g	$g(\lambda_1)$	$g(\lambda_2)$	$\cdots$	$g(\lambda_m)$

The matrices $P_{\lambda_i} := f_{\lambda_i}(A)$ are the **spectral projections**.

The Value Table: ID Card for $g(A)$

For our A with eigenvalues $0, 1, 2$:

every $g(A)$ is determined by three numbers $\big(g(0),\ g(1),\ g(2)\big)$.

$g(t)$	$g(0)$	$g(1)$	$g(2)$	$g(A)$
1	1	1	1	I
t	0	1	2	A
t^2	0	1	4	A^2
t^n	0	1	2^n	A^n

The Value Table: ID Card for $g(A)$

For our A with eigenvalues $0, 1, 2$:

every $g(A)$ is determined by three numbers $\big(g(0),\ g(1),\ g(2)\big)$.

$g(t)$	$g(0)$	$g(1)$	$g(2)$	$g(A)$
1	1	1	1	I
t	0	1	2	A
t^2	0	1	4	A^2
t^n	0	1	2^n	A^n

The value table is the **complete data** of $g(A)$.

Two polynomials with the **same value table** give the **same matrix**.

Arithmetic in Value Tables

Addition: value tables add entry-wise.

$$\underbrace{(0,1,2)}_{A} + \underbrace{(1,1,1)}_{I} = \underbrace{(1,2,3)}_{A+I} \qquad \checkmark$$

Arithmetic in Value Tables

Addition: value tables add entry-wise.

$$\underbrace{(0,1,2)}_{A} + \underbrace{(1,1,1)}_{I} = \underbrace{(1,2,3)}_{A+I} \qquad \checkmark$$

Multiplication: value tables **multiply entry-wise**!

$$\underbrace{(0,1,2)}_{A} \cdot \underbrace{(0,1,2)}_{A} = \underbrace{(0,1,4)}_{A^2} \qquad \checkmark$$

Why? Because $g(A) \cdot h(A) = (g \cdot h)(A)$, and $(g \cdot h)(\lambda_i) = g(\lambda_i) \cdot h(\lambda_i)$.

Arithmetic in Value Tables

Addition: value tables add entry-wise.

$$\underbrace{(0,1,2)}_{A} + \underbrace{(1,1,1)}_{I} = \underbrace{(1,2,3)}_{A+I} \qquad \checkmark$$

Multiplication: value tables **multiply entry-wise**!

$$\underbrace{(0,1,2)}_{A} \cdot \underbrace{(0,1,2)}_{A} = \underbrace{(0,1,4)}_{A^2} \qquad \checkmark$$

Why? Because $g(A) \cdot h(A) = (g \cdot h)(A)$, and $(g \cdot h)(\lambda_i) = g(\lambda_i) \cdot h(\lambda_i)$.

Matrix multiplication (hard) $\longleftrightarrow$ entry-wise multiplication (easy).

The value table turns matrix algebra into $\mathbb{R}^3$ **arithmetic**.

The Spectral Projections = Standard Basis

Matrix	$t = 0$	$t = 1$	$t = 2$
$P_0 = f_0(A)$	**1**	0	0
$P_1 = f_1(A)$	0	**1**	0
$P_2 = f_2(A)$	0	0	**1**

These are the **standard basis vectors** of value-table space!

The Spectral Projections = Standard Basis

Matrix	$t = 0$	$t = 1$	$t = 2$
$P_0 = f_0(A)$	**1**	0	0
$P_1 = f_1(A)$	0	**1**	0
$P_2 = f_2(A)$	0	0	**1**

These are the **standard basis vectors** of value-table space!

Any value table (a, b, c) decomposes as:

$$(a, b, c) = a \cdot (1, 0, 0) + b \cdot (0, 1, 0) + c \cdot (0, 0, 1)$$

So: $g(A) = g(0) \cdot P_0 + g(1) \cdot P_1 + g(2) \cdot P_2$.

The spectral projections play the role of **coordinate axes**.

Testing Properties with Value Tables

Is P_1 a projection? Value table: $(0, 1, 0)$.

Square entry-wise: $(0^2, 1^2, 0^2) = (0, 1, 0)$. Same! $P_1^2 = P_1$ ✓

Testing Properties with Value Tables

Is P_1 a projection? Value table: $(0, 1, 0)$.

Square entry-wise: $(0^2, 1^2, 0^2) = (0, 1, 0)$. Same! $P_1^2 = P_1$ ✓

Is A a projection? Value table: $(0, 1, 2)$.

Square entry-wise: $(0^2, 1^2, 2^2) = (0, 1, 4) \neq (0, 1, 2)$. $A^2 \neq A$ ×

Testing Properties with Value Tables

Is P_1 a projection? Value table: $(0, 1, 0)$.

Square entry-wise: $(0^2, 1^2, 0^2) = (0, 1, 0)$. Same! $P_1^2 = P_1$ ✓

Is A a projection? Value table: $(0, 1, 2)$.

Square entry-wise: $(0^2, 1^2, 2^2) = (0, 1, 4) \neq (0, 1, 2)$. $A^2 \neq A$ ×

Is $P_0 + P_1$ a projection? Value table: $(1, 1, 0) + (0, 1, 0)\ldots$

Wait — that gives $(1, 2, 0)$, and $2^2 = 4 \neq 2$. Not a projection.

Testing Properties with Value Tables

Is P_1 a projection? Value table: $(0, 1, 0)$.

Square entry-wise: $(0^2, 1^2, 0^2) = (0, 1, 0)$. Same! $P_1^2 = P_1$ ✓

Is A a projection? Value table: $(0, 1, 2)$.

Square entry-wise: $(0^2, 1^2, 2^2) = (0, 1, 4) \neq (0, 1, 2)$. $A^2 \neq A$ ×

Is $P_0 + P_1$ a projection? Value table: $(1, 1, 0) + (0, 1, 0)\ldots$

Wait — that gives $(1, 2, 0)$, and $2^2 = 4 \neq 2$. Not a projection.

Is $P_0 + P_2$ a projection? Value table: $(1, 0, 0) + (0, 0, 1) = (1, 0, 1)$.

Square: $(1, 0, 1)$. Same! $(P_0 + P_2)^2 = P_0 + P_2$ ✓

Rule: $g(A)$ is a projection $\iff$ every entry of its value table is 0 **or** 1.

Compatible Projections — Instantly

Are $\{P_0, P_1, P_2\}$ a **compatible family** (from Lecture 9)?

Need: $P_i P_j = 0$ for $i \neq j$. Multiply value tables entry-wise:

$$P_0 \cdot P_1 : \quad (1,0,0) \cdot (0,1,0) = (0,0,0) \quad \Longrightarrow \quad P_0 P_1 = 0 \checkmark$$

$$P_0 \cdot P_2 : \quad (1,0,0) \cdot (0,0,1) = (0,0,0) \quad \Longrightarrow \quad P_0 P_2 = 0 \checkmark$$

$$P_1 \cdot P_2 : \quad (0,1,0) \cdot (0,0,1) = (0,0,0) \quad \Longrightarrow \quad P_1 P_2 = 0 \checkmark$$

Compatible Projections — Instantly

Are $\{P_0, P_1, P_2\}$ a **compatible family** (from Lecture 9)?

Need: $P_i P_j = 0$ for $i \neq j$. Multiply value tables entry-wise:

$$P_0 \cdot P_1 : \quad (1,0,0) \cdot (0,1,0) = (0,0,0) \quad \Longrightarrow \quad P_0 P_1 = 0 \checkmark$$

$$P_0 \cdot P_2 : \quad (1,0,0) \cdot (0,0,1) = (0,0,0) \quad \Longrightarrow \quad P_0 P_2 = 0 \checkmark$$

$$P_1 \cdot P_2 : \quad (0,1,0) \cdot (0,0,1) = (0,0,0) \quad \Longrightarrow \quad P_1 P_2 = 0 \checkmark$$

Compatible family — no matrix multiplication needed!

And $P_0 + P_1 + P_2$: value table $(1,0,0) + (0,1,0) + (0,0,1) = (1,1,1) =$ value table of I.

$$\boxed{P_0 + P_1 + P_2 = I}$$

The projections **partition the identity** — just like Lectures 7–9!

Spectral Decomposition: The Dream Comes True

What has value table $(0, 1, 2)$?

$$0 \cdot \underbrace{(1,0,0)}_{P_0} + 1 \cdot \underbrace{(0,1,0)}_{P_1} + 2 \cdot \underbrace{(0,0,1)}_{P_2} = (0,1,2) = \text{value table of } A$$

$$\boxed{A = 0 \cdot P_0 + 1 \cdot P_1 + 2 \cdot P_2 = P_1 + 2P_2}$$

Spectral Decomposition: The Dream Comes True

What has value table $(0, 1, 2)$?

$$0 \cdot \underbrace{(1,0,0)}_{P_0} + 1 \cdot \underbrace{(0,1,0)}_{P_1} + 2 \cdot \underbrace{(0,0,1)}_{P_2} = (0,1,2) = \text{value table of } A$$

$$\boxed{A = 0 \cdot P_0 + 1 \cdot P_1 + 2 \cdot P_2 = P_1 + 2P_2}$$

Verify with the concrete matrices:

$$1 \cdot \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + 2 \cdot \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = A \checkmark$$

The **dream from §1** is real: $A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_m P_m$.

$(A - \lambda_i I) \cdot P_{\lambda_i} = 0$ — by Value Tables

$\lambda_1 = 0$: value table of $(A - 0 \cdot I) = A$ is $(0, 1, 2)$.

$$(0, 1, 2) \cdot \underbrace{(1, 0, 0)}_{P_0} = (0, 0, 0) = 0 \quad \checkmark$$

$(A - \lambda_i I) \cdot P_{\lambda_i} = 0$ — by Value Tables

$\lambda_1 = 0$: value table of $(A - 0 \cdot I) = A$ is $(0, 1, 2)$.

$$(0, 1, 2) \cdot \underbrace{(1, 0, 0)}_{P_0} = (0, 0, 0) = 0 \ \checkmark$$

$\lambda_2 = 1$: value table of $(A - I)$ is $(0-1,\ 1-1,\ 2-1) = (-1, 0, 1)$.

$$(-1, 0, 1) \cdot \underbrace{(0, 1, 0)}_{P_1} = (0, 0, 0) = 0 \ \checkmark$$

$(A - \lambda_i I) \cdot P_{\lambda_i} = 0$ — by Value Tables

$\lambda_1 = 0$: value table of $(A - 0 \cdot I) = A$ is $(0, 1, 2)$.

$$(0, 1, 2) \cdot \underbrace{(1, 0, 0)}_{P_0} = (0, 0, 0) = 0 \ \checkmark$$

$\lambda_2 = 1$: value table of $(A - I)$ is $(0-1,\ 1-1,\ 2-1) = (-1, 0, 1)$.

$$(-1, 0, 1) \cdot \underbrace{(0, 1, 0)}_{P_1} = (0, 0, 0) = 0 \ \checkmark$$

$\lambda_3 = 2$: value table of $(A - 2I)$ is $(-2, -1, 0)$.

$$(-2, -1, 0) \cdot \underbrace{(0, 0, 1)}_{P_2} = (0, 0, 0) = 0 \ \checkmark$$

Why it works: $(t - \lambda_i)$ is zero at λ_i, P_{λ_i} is zero everywhere **else**.

Their product is zero at **every** eigenvalue $\implies (A - \lambda_i I)\, P_{\lambda_i} = 0$.

Eigenvectors: Just Read the Columns

We just showed: $(A - \lambda_i I)\, P_{\lambda_i} = 0$, i.e. $A\, P_{\lambda_i} = \lambda_i\, P_{\lambda_i}$.

Column by column: every **non-zero column** of P_{λ_i} is an **eigenvector** for λ_i.

λ	P_{λ_i}	Eigenvector	Verify
0	$\frac{1}{2}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$	$A\vec{v} = \vec{0} = 0\vec{v}$ ✓
1	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$	$A\vec{v} = \vec{v} = 1\vec{v}$ ✓
2	$\frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$	$A\vec{v} = 2\vec{v}$ ✓

No equation solving. No Gaussian elimination. Just **read the columns**.

Full Example: The Value Table Method from Scratch

New Example: Find A^n

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Characteristic polynomial:

$$\det(tI - A) = (t-1)(t-3)(t-5)$$

Eigenvalues: $\lambda_1 = 1, \quad \lambda_2 = 3, \quad \lambda_3 = 5.$

New Example: Find A^n

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Characteristic polynomial:

$$\det(tI - A) = (t-1)(t-3)(t-5)$$

Eigenvalues: $\lambda_1 = 1, \quad \lambda_2 = 3, \quad \lambda_3 = 5.$

Goal: find A^n using value tables.

Step 1: Build the Value Tables

Evaluate each matrix polynomial at the eigenvalues $1, 3, 5$:

$g(t)$	$g(1)$	$g(3)$	$g(5)$	$g(A)$
1	1	1	1	$= I$
t	1	3	5	$= A$
t^2	1	9	25	$= A^2$
t^n	1	3^n	5^n	$= A^n$ **(this is what we want!)**

Step 1: Build the Value Tables

Evaluate each matrix polynomial at the eigenvalues $1, 3, 5$:

$g(t)$	$g(1)$	$g(3)$	$g(5)$	$g(A)$
1	1	1	1	$= I$
t	1	3	5	$= A$
t^2	1	9	25	$= A^2$
t^n	1	3^n	5^n	$= A^n$ **(this is what we want!)**

We need the **projections** P_1, P_3, P_5 with value tables

$$(1, 0, 0), \qquad (0, 1, 0), \qquad (0, 0, 1)$$

Then $A^n = 1^n \cdot P_1 + 3^n \cdot P_3 + 5^n \cdot P_5$.

Step 2: Find Polynomials That Realize the Value Tables

Goal: find polynomials with value tables $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.

	$\underbrace{\frac{(t-3)(t-5)}{8}}_{P_1}$	$\underbrace{\frac{(t-1)(t-5)}{-4}}_{P_3}$	$\underbrace{\frac{(t-1)(t-3)}{8}}_{P_5}$
$t = 1$	$\frac{(-2)(-4)}{8} = \mathbf{1}$	0	0
$t = 3$	0	$\frac{(2)(-2)}{-4} = \mathbf{1}$	0
$t = 5$	0	0	$\frac{(4)(2)}{8} = \mathbf{1}$

Step 2: Find Polynomials That Realize the Value Tables

Goal: find polynomials with value tables $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.

	$\underbrace{\frac{(t-3)(t-5)}{8}}_{P_1}$	$\underbrace{\frac{(t-1)(t-5)}{-4}}_{P_3}$	$\underbrace{\frac{(t-1)(t-3)}{8}}_{P_5}$
$t=1$	$\frac{(-2)(-4)}{8} = \mathbf{1}$	0	0
$t=3$	0	$\frac{(2)(-2)}{-4} = \mathbf{1}$	0
$t=5$	0	0	$\frac{(4)(2)}{8} = \mathbf{1}$

Value tables match the targets! Plug in $t = A$:

$$P_1 = \frac{(A-3I)(A-5I)}{8} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$P_3 = \frac{(A-I)(A-5I)}{-4} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$P_5 = \frac{(A-I)(A-3I)}{8} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Why Lagrange? It builds polynomials that realize target value tables.
That's its only job.

Alternative: Find Projections by Solving a System

Forgot the Lagrange formula? The **value table is the goal** — Lagrange is just one tool.

Alternative: write $P_1 = \alpha_1 I + \alpha_2 A + \alpha_3 A^2$ and match value tables:

$$\alpha_1 \underbrace{(1,1,1)}_{I} + \alpha_2 \underbrace{(1,3,5)}_{A} + \alpha_3 \underbrace{(1,9,25)}_{A^2} = (\mathbf{1}, 0, 0)$$

Entry by entry — a 3×3 system:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \Longrightarrow \qquad \alpha_1 = \tfrac{15}{8}, \quad \alpha_2 = -1, \quad \alpha_3 = \tfrac{1}{8}$$

Alternative: Find Projections by Solving a System

Forgot the Lagrange formula? The **value table is the goal** — Lagrange is just one tool.

Alternative: write $P_1 = \alpha_1 I + \alpha_2 A + \alpha_3 A^2$ and match value tables:

$$\alpha_1 \underbrace{(1,1,1)}_{I} + \alpha_2 \underbrace{(1,3,5)}_{A} + \alpha_3 \underbrace{(1,9,25)}_{A^2} = (\mathbf{1},0,0)$$

Entry by entry — a 3×3 system:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \Longrightarrow \qquad \alpha_1 = \tfrac{15}{8}, \;\; \alpha_2 = -1, \;\; \alpha_3 = \tfrac{1}{8}$$

$$P_1 = \tfrac{15}{8}\, I - A + \tfrac{1}{8}\, A^2 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \checkmark \quad \text{Same answer!}$$

The **value table determines the matrix**. How you find the polynomial doesn't matter.

Quick Check: Value Table Tests

Projections? Square each value table entry-wise:

$$
\begin{aligned}
P_1: &\quad (1,0,0)^2 = (1,0,0) &&\implies \quad P_1^2 = P_1 \checkmark \\
P_3: &\quad (0,1,0)^2 = (0,1,0) &&\implies \quad P_3^2 = P_3 \checkmark \\
P_5: &\quad (0,0,1)^2 = (0,0,1) &&\implies \quad P_5^2 = P_5 \checkmark
\end{aligned}
$$

Compatible? Multiply value tables entry-wise:

$$
\begin{aligned}
(1,0,0)\cdot(0,1,0) = (0,0,0) &\quad \implies \quad P_1 P_3 = 0 \checkmark \\
(1,0,0)\cdot(0,0,1) = (0,0,0) &\quad \implies \quad P_1 P_5 = 0 \checkmark \\
(0,1,0)\cdot(0,0,1) = (0,0,0) &\quad \implies \quad P_3 P_5 = 0 \checkmark
\end{aligned}
$$

Partition of I? $(1,0,0) + (0,1,0) + (0,0,1) = (1,1,1)$

$$\implies \quad P_1 + P_3 + P_5 = I$$

Quick Check: Value Table Tests

Projections? Square each value table entry-wise:

$$P_1: \quad (1,0,0)^2 = (1,0,0) \quad \Longrightarrow \quad P_1^2 = P_1 \checkmark$$
$$P_3: \quad (0,1,0)^2 = (0,1,0) \quad \Longrightarrow \quad P_3^2 = P_3 \checkmark$$
$$P_5: \quad (0,0,1)^2 = (0,0,1) \quad \Longrightarrow \quad P_5^2 = P_5 \checkmark$$

Compatible? Multiply value tables entry-wise:

$$(1,0,0)\cdot(0,1,0) = (0,0,0) \quad \Longrightarrow \quad P_1 P_3 = 0 \checkmark$$
$$(1,0,0)\cdot(0,0,1) = (0,0,0) \quad \Longrightarrow \quad P_1 P_5 = 0 \checkmark$$
$$(0,1,0)\cdot(0,0,1) = (0,0,0) \quad \Longrightarrow \quad P_3 P_5 = 0 \checkmark$$

Partition of I? $(1,0,0) + (0,1,0) + (0,0,1) = (1,1,1)$

$$\Longrightarrow \quad P_1 + P_3 + P_5 = I$$

All verified — without any matrix multiplication.

Step 3: A^n by Value Tables

A^n has value table $(1^n,\ 3^n,\ 5^n) = (1,\ 3^n,\ 5^n)$.

Express in the standard basis:

$$(1,\ 3^n,\ 5^n) \ = \ 1 \cdot \underbrace{(1,0,0)}_{P_1} + 3^n \cdot \underbrace{(0,1,0)}_{P_3} + 5^n \cdot \underbrace{(0,0,1)}_{P_5}$$

Value tables match $\implies$ matrices are equal:

$$\boxed{A^n = P_1 + 3^n\, P_3 + 5^n\, P_5 = \begin{pmatrix} 1 & 3^n - 1 & 0 \\ 0 & 3^n & 0 \\ 0 & 0 & 5^n \end{pmatrix}}$$

Step 3: A^n by Value Tables

A^n has value table $(1^n,\ 3^n,\ 5^n) = (1,\ 3^n,\ 5^n)$.

Express in the standard basis:

$$(1,\ 3^n,\ 5^n) \ = \ 1 \cdot \underbrace{(1,0,0)}_{P_1} + 3^n \cdot \underbrace{(0,1,0)}_{P_3} + 5^n \cdot \underbrace{(0,0,1)}_{P_5}$$

Value tables match $\implies$ matrices are equal:

$$\boxed{A^n = P_1 + 3^n\, P_3 + 5^n\, P_5 = \begin{pmatrix} 1 & 3^n - 1 & 0 \\ 0 & 3^n & 0 \\ 0 & 0 & 5^n \end{pmatrix}}$$

Check: $A^1 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ ✓ $\qquad A^2 = \begin{pmatrix} 1 & 8 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 25 \end{pmatrix}$ ✓

Now let's solve the problem from the beginning.

What is a_{100}?

Solving the Recurrence: $a_{n+1} = 3a_n - 2a_{n-1}$

$B = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$, $\det(tI - B) = (t-1)(t-2)$. Eigenvalues: $1, 2$.

Projections by value tables:

P_1: value table $(1, 0)$, zero at $t = 2$:

$$P_1 = \frac{-(B - 2I)}{1 - 2} = -(B - 2I) = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$$

P_2: value table $(0, 1)$, zero at $t = 1$: $P_2 = \dfrac{B - I}{2 - 1} = B - I = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$

B^n by value tables: $(1^n, 2^n) = 1 \cdot (1, 0) + 2^n \cdot (0, 1)$, so $B^n = P_1 + 2^n P_2$.

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = B^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} + 2^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2^{n+1} - 1 \\ 2^n - 1 \end{pmatrix}$$

$$\boxed{a_n = 2^n - 1}$$

Check: $a_0 = 0$, $a_1 = 1$, $a_2 = 3$, $a_3 = 7$. ✓

Summary: What Is Spectral Decomposition?

If $\det(tI - A) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_m)$ with distinct λ_i, then:

$$\boxed{A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_m P_m}$$

where $P_1, \ldots, P_m$ are **compatible projections** that **partition** I.

Summary: What Is Spectral Decomposition?

If $\det(tI - A) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_m)$ with distinct λ_i, then:

$$\boxed{A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_m P_m}$$

where $P_1, \ldots, P_m$ are **compatible projections** that **partition** I.

Why does the value table method work?

Any polynomial $g(A)$ decomposes as:

$$g(A) = \underbrace{g(\lambda_1)}_{\text{value at } \lambda_1} \cdot P_1 + \underbrace{g(\lambda_2)}_{\text{value at } \lambda_2} \cdot P_2 + \cdots + \underbrace{g(\lambda_m)}_{\text{value at } \lambda_m} \cdot P_m$$

The projections P_i are **fixed** — they depend only on A, not on g.

The value table $\big(g(\lambda_1), \ldots, g(\lambda_m)\big)$ gives the **coefficients**

for the linear combination of these projections.

Same value table $\implies$ same coefficients $\implies$ **same matrix**.

That is the entire reason the value table method works.

Next Lecture

Today we found the spectral projections $P_1, \ldots, P_m$ as **polynomials of A**.

But what ARE these projections, geometrically?

Next Lecture

Today we found the spectral projections $P_1, \ldots, P_m$ as **polynomials of A**.

But what ARE these projections, geometrically?

- What **subspace** does each P_i project onto?
- These subspaces are the **eigenspaces** of A.
- When the eigenspaces span $\mathbb{R}^n$: **diagonalization**.

Next time: **Eigenspaces and Diagonalization**.